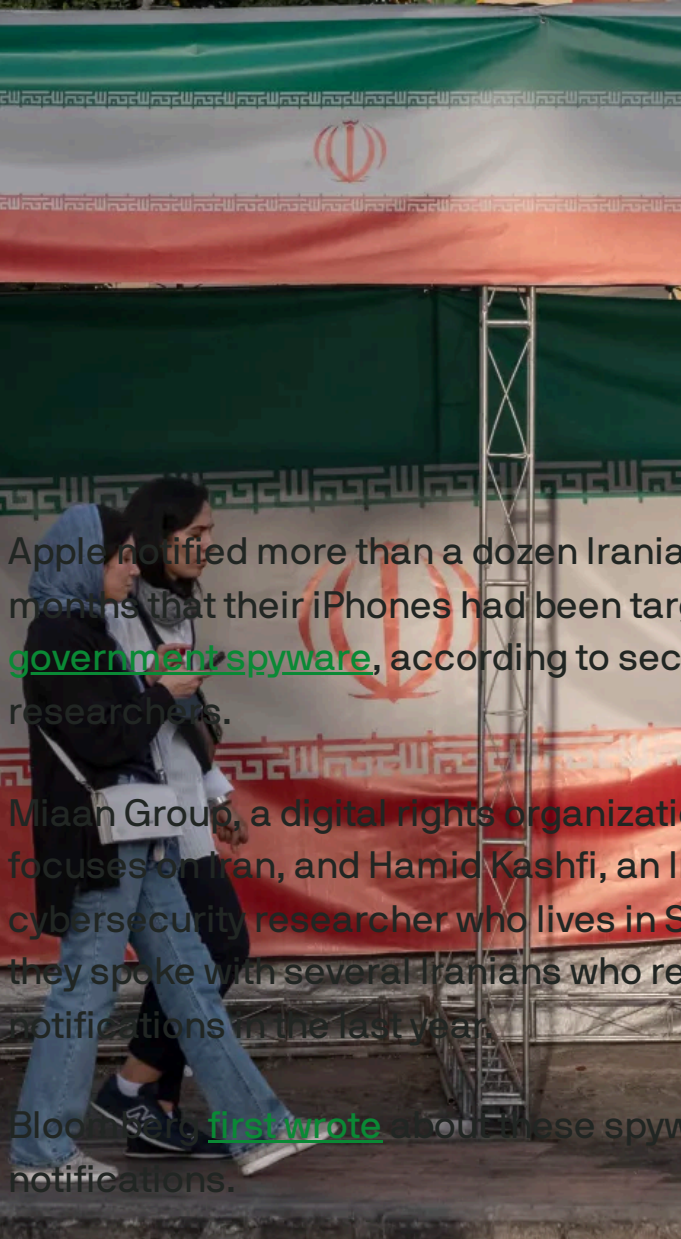


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SECURITY



Apple alerted Iranians to iPhone spyware attacks, say researchers



Lorenzo Franceschi-
Bicchierai

— 1:57 PM PDT · July 22,
2025

Apple notified more than a dozen Iranians in recent months that their iPhones had been targeted with **government spyware**, according to security researchers.

Miaan Group, a digital rights organization that focuses on Iran, and Hamid Kashfi, an Iranian cybersecurity researcher who lives in Sweden, said they spoke with several Iranians who received the notifications in the last year.

Bloomberg **first wrote** about these spyware notifications.

IMAGE CREDITS:

MORTEZA NIKOUBAZEL/NORPHOTO / GETTY IMAGES

Miaan Group **published a report** on Tuesday on the state of cybersecurity of civil society in Iran, which mentioned that the organization's researchers have identified three cases of government spyware attacks against Iranians, two in Iran and one in Europe, who were alerted in April of this year.

"Two people in Iran come from a family with a long history of political activism against the Islamic Republic. Many members of their family have been executed, and they have no history of traveling abroad," Amir Rashidi, Miaan Group's director of digital rights and security, told TechCrunch. "I believe there have been three waves of attacks, and we have only seen the tip of the iceberg."

Rashidi said that Iran is likely the government behind the attacks, although there needs to be

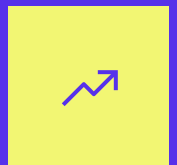
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more investigations into these attacks to reach a more conclusive determination. “I see no reason for members of civil society to be targeted by anyone other than Iran,” he said.

Kashfi, who founded the security firm DarkCell, said in an email that he helped two victims go through preliminary forensics steps, but he wasn’t able to confirm which spyware maker was behind the attacks. And, he added, some of the victims he worked with preferred not to continue the investigation.

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Have you received a threat notification from Apple? We’d love to hear from you. From a non-work device and network, you can contact Lorenzo Franceschi-Bicchierai securely on Signal at +1 917 257 1382, or via Telegram and Keybase @lorenzofb, or [email](#).

”Pretty much all victims spooked out and ghosted us as soon as we explained the seriousness of the case to them. I presume partly because of their place of work and sensitivity of the matters related to that,” said Kashfi, who added that one of the victims received the notification in 2024.

It’s unclear which spyware maker is behind these attacks.

Over the last few years, Apple has sent [several rounds](#) of [notifications](#) to people whom the company believes have been targeted with government spyware, such as [NSO Group’s Pegasus](#), or [Paragon’s Graphite](#). This kind of malware is also known as “mercenary” or “commercial” spyware.

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The notifications have helped security researchers who focus on spyware to document abuses in several countries such as [India](#), [El Salvador](#), and [Thailand](#).

On [Apple's support page](#) for what the company calls “threat notifications,” last updated in April, the tech giant said that since 2021 it has notified users in “in over 150 countries,” which shows how widespread the use of government spyware is. Apple does not disclose the names of the countries, nor the total number of people it has notified.

To help victims, since last year, Apple [has recommended those who received these threat notifications to reach out to digital rights group AccessNow](#), which runs an around-the-clock helpline staffed with researchers who can investigate spyware attacks. AccessNow has documented cases of spyware abuse all over the world.

Apple did not respond to a request for comment on the notifications sent to Iranians.

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IMAGE CREDITS: BRYCE DURBIN / TECHCRUNCH

SECURITY



Apple sends spyware victims to this nonprofit security lab

Lorenzo Franceschi-Bicchierai

6:13 AM PST · December 20, 2024

Before the elections, the cybersecurity team of U.S. vice president and then-presidential candidate Kamala Harris reached out to Apple asking for help, [according to Forbes](#), after a tool that's designed to detect spyware on iPhones flagged anomalies on two devices belonging to campaign staffers. Apple declined to forensically analyze the phones, per Forbes.

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The company’s response is no surprise to the digital defenders working with at-risk populations often targeted by spyware.

In the last few years, Apple has been [sending notifications](#) to targets and victims of government spyware, alerting them that they may have been hacked, [and directing them to get help](#). Crucially, Apple doesn’t tell the targets to get in touch with its own security engineers, but with the [nonprofit Access Now](#), which runs a digital helpline for people in civil society who suspect they have been targets of government spyware.

“Apple detected that you are being targeted by a mercenary spyware attack that is trying to remotely compromise the iPhone associated with your Apple Account,” reads a recent alert, which Access Now shared with TechCrunch. “This attack is likely targeting you specifically because of who you are or what you do. Although it’s never possible to achieve absolute certainty when detecting such attacks, Apple has high confidence in this warning — please take it seriously.”

While it may look like Apple is abdicating its responsibility to protect its users, cybersecurity experts who work with human rights defenders, journalists, and dissidents generally agree that Apple’s approach in alerting victims to spyware attacks is the right one.

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Do you have more information about government spyware and its makers? From a non-work device, you can contact Lorenzo Franceschi-Bicchierai securely on Signal at +1 917 257 1382, or via Telegram and Keybase [@lorenzofb](#), or

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“These notifications have been a game changer for spyware accountability research,” said John Scott-Railton, a senior researcher at the Citizen Lab, a nonprofit that investigates spyware and is housed at the University of Toronto Munk School of Global Affairs & Public Policy.

“When I look back over the past few years, I see so many of the most important cases that we know about — [Poland](#), [Thailand](#), so many [others](#) — began with an Apple notification,” said Scott-Railton.

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For people who investigate spyware, Apple sharing spyware notifications with victims represented a turning point. Before the notifications, “We were just like in the dark, not knowing who to check,” according to Access Now’s legal counsel Natalia Krapiva.

“I think it’s one of the greatest things that’s happened in the sphere of this kind of forensic investigations and hunting of sophisticated spyware,” Krapiva told TechCrunch.

Now, when someone or a group of people get a notification from Apple, they are warned that something potentially anomalous is happening with their device, that someone is targeting them, and that they need to get help. And Apple tells them exactly where to get it, according to Scott-Railton, who said Access Now’s helpline is the right place to go because “the helpline is able to do good, systematic triage work and support.”

Krapiva said that the helpline is staffed by more than 30 people, supported by others who work in other departments of the nonprofit. So far in 2024, Krapiva said Access Now received 4,337 tickets through the helpline.

Scott-Railton, Krapiva, and security expert Runa Sandvik, [who runs her own digital security consultancy Granitt](#) for at-risk people and has been protecting journalists for a decade, all agree Apple should stop short of investigating individual attacks after notifying the victims.

“Big tech companies don’t want to get into the business of doing forensics on people’s devices or

accounts,” Sandvik told TechCrunch. “I think that should remain separate.”

Eva Galperin, the director of cybersecurity at the nonprofit Electronic Frontier Foundation, who has been investigating surveillance on the internet for more than a decade, said that Apple could still do more to combat spyware.

“[Apple] could write more detailed reports and file more lawsuits. These are the things that take massive amounts of money NGOs don’t have and telemetry NGOs don’t have,” Galperin told TechCrunch.

[In its official page](#) about mercenary spyware, last updated in October, Apple says that since 2012 it has sent notifications to users in more than 150 countries.

Apple spokesperson Nadine Haija told TechCrunch that the “vast majority of users will never be the victims of such attacks, we sympathize deeply with the small number of users who are, and we continue to work tirelessly to protect them,” and reiterated that there are no known cases of mercenary spyware on Apple devices with Lockdown Mode. “Our security teams are constantly working to track mercenary spyware attackers, and we send threat notifications to inform and assist users who we believe were individually targeted.”

For anyone alerted by a notification, Apple tells those targets and victims of spyware to update their iOS software and all their apps. Apple also suggests the user switches on Lockdown Mode, an opt-in iOS security feature [that has stopped spyware attacks in the past](#) by limiting device features that are often exploited to plant spyware. Apple [said last year that it is not aware of any successful spyware infection](#) against someone who used Lockdown Mode.

Scott-Railton called Lockdown Mode “a game changer in increasing the security of people’s devices, especially people who are at risk.”

All the experts TechCrunch spoke with [strongly recommend turning on Lockdown Mode](#) if you think you may be a target, especially if you are a journalist, human rights defender, or dissident.

And if you get a notification from Apple, take it very seriously.

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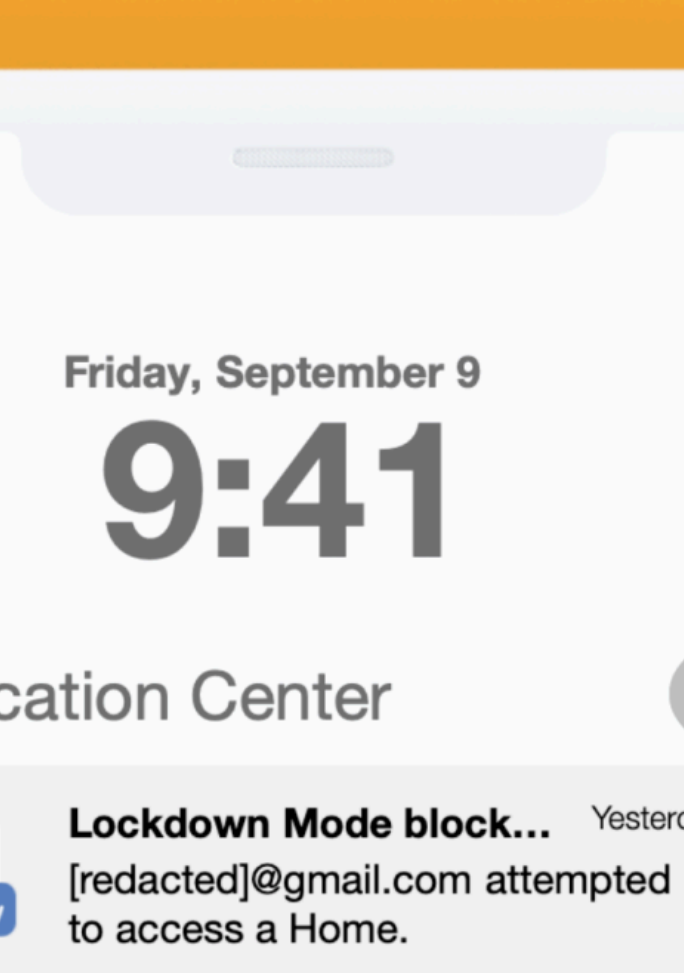
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Apple's high security mode blocked NSO spyware, researchers say

Lorenzo Franceschi-Bicchierai

— 4:11 AM PDT · April 18, 2023

Lockdown Mode block... Yesterday
[redacted]@gmail.com attempted to access a Home.

Lockdown Mode block... M
[redacted]@yahoo.com attempted to access a Home.

Last year, Apple [launched a new feature](#) for iPhone users who are worried about getting targeted with sophisticated spyware, such as journalists or human rights defenders. Now, researchers say they have found evidence that the feature — [called Lockdown Mode](#) — helped block an attack by hackers using spyware made by the infamous mercenary hacking provider NSO Group.

On Tuesday, the cybersecurity and human rights research group Citizen Lab [released a report](#) analyzing three new zero-day exploits in iOS 15 and iOS 16 — meaning Apple was unaware of the vulnerabilities at the time they were [used to target](#) at least two Mexican human rights defenders.

One of those exploits was blocked by Lockdown Mode, the researchers found. Lockdown Mode was specifically designed to reduce the iPhone's attack surface — cybersecurity lingo referring to parts of the code or features of a system prone to attacks by

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hackers. This is the first documented case where Lockdown Mode has successfully protected someone from a targeted attack.

In the recent cases, Citizen Lab researchers said that the targets' iPhones blocked the hacking attempts and showed a notification saying Lockdown Mode prevented someone from accessing the phone's Home app. The researchers, however, note that it's possible that at some point NSO's exploit developers "may have figured out a way to correct the notification issue, such as by fingerprinting Lockdown Mode."

As other researchers have pointed out in the past, [it's easy to fingerprint users to determine who has Lockdown Mode turned on](#), but that's not to say its protections are not meaningful. As this case found by Citizen Lab shows, Lockdown Mode can be effective.

"The fact that Lockdown Mode seems to have thwarted, and even notified targets of a real-world zero-click attack shows that it is a powerful mitigation, and is a cause for great optimism," Bill Marczak, a senior researcher at Citizen Lab and one of the authors of the report, told TechCrunch. "But, as with any optional feature, the devil is always in the details. How many people will opt to turn on Lockdown Mode? Will attackers simply move away from exploiting Apple apps and target third-party apps, which are harder for Lockdown Mode to secure?"

Apple spokesperson Scott Radcliffe said in a statement: "We are pleased to see that Lockdown Mode disrupted this sophisticated attack and alerted users immediately, even before the specific threat was known to Apple and security researchers. Our security teams around the world will continue to work tirelessly to advance

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Lockdown Mode and strengthen the security and privacy protections in iOS.”

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NSO Group spokesperson Liron Bruck did not respond to a series of questions, instead sending a statement saying that “Citizen Lab has repeatedly produced reports that are unable to determine the technology in use and they refuse to share their underlying data. NSO adheres to strict regulation and its technology is used by its governmental

customers to fight terror and crime around the world.”

Citizen Lab’s report identified three different exploits — all “zero-click,” meaning they did not require any interaction by the target — by analyzing several phones that were suspected to have been hacked with NSO’s spyware, also known as Pegasus.

Pegasus, which NSO sells exclusively to government customers, can remotely obtain a phone’s location, messages, photos and virtually anything the phone’s legitimate owner can access. For years, researchers at Citizen Lab, Amnesty International and other organizations have documented several cases where NSO customers used the company’s spyware to target [journalists](#), [human rights defenders](#) and [opposition politicians](#).

Citizen Lab’s new findings show that NSO is still alive and well, despite a rocky past couple of years. In 2021, an international consortium of media organizations [launched the Pegasus Project](#), a series of articles detailing scandals involving NSO all over the world. Then, later that year, the U.S. government put NSO on [a denylist](#), effectively barring any U.S. company or individual from doing business with the company.

“Other companies have folded, but, at least for now, NSO is still able to bear these increased costs, and Pegasus remains an active threat to global civil society,” Marczak said.

Of the recent batch of exploits: The first exploit was deployed in January 2022 by NSO customers and exploited the iPhone’s FindMy feature, which helps owners locate their lost or stolen phones. The second exploit deployed starting in June 2022 and is a “two-step” exploit, meaning it targets two features, in this case the FindMy feature and

iMessage. And the last exploit, deployed starting in October 2022 exploited the iPhone's HomeKit and iMessage functionalities.

In its report, Citizen Lab said the two Mexicans targeted by the exploits investigate human rights violations allegedly carried out by the Mexican military. The Mexican government is a [known spyware customer](#).

Citizen Lab reported all these exploits to Apple, which have since then pushed updates and reduced the attack surface. Apple fixed the HomeKit-based vulnerability in iOS 16.3.1, released in February.

Do you have more information about NSO Group? Or another surveillance tech provider? We'd love to hear from you. You can contact Lorenzo Franceschi-Bicchierai securely on Signal at +1 917 257 1382, or via Wickr, Telegram and Wire @lorenzofb, or email lorenzo@techcrunch.com. You can also contact TechCrunch via [SecureDrop](#).

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